Lesson 1.2 - Descriptive Statistics vs Inferential Statistics

> **Video:** Descriptive Statistics vs Inferential Statistics by The Organic Chemistry Tutor **Chapter:** 1 - Introduction to Statistics

What is statistics?

Statistics is the science of **collecting, analyzing, interpreting and presenting data**. It has two main branches:

	Descriptive statistics	Inferential statistics
Goal	**Organize and summarize** the data you have	**Draw conclusions about a population** from a sample
Tools	Graphs, tables, summary numbers	Probability, estimates, confidence intervals, hypothesis tests
Typical statement	"20 of the 100 people surveyed said yes."	"About 20% of the whole city would say yes, ± 2%."

You usually learn descriptive statistics first, because inference is built on top of it.

Descriptive statistics

Descriptive statistics describes data in two ways.

1. With graphs and tables

- **Bar graph:** separate bars, one per category.
- **Histogram:** like a bar graph, but the bars **touch** because they cover consecutive numeric ranges.
- **Pie chart:** shows how a whole is split into percentages.
- **Line graph:** shows change, often over time.
- **Frequency table / frequency distribution table:** counts how often each value or class occurs.
- **Shape of the data:** symmetric, skewed right, or skewed left.

2. With numbers

- **Measures of central tendency:**
  - **Mean** (average) = sum of values ÷ number of values.

        - *Sample mean* $\bar{x}$: averages the values in a sample.
        - *Population mean* μ: averages every value in the population.
    - **Median**: the middle value of the sorted data.
    - **Mode**: the most frequent value.
- **Measures of spread:** range, variance, standard deviation.

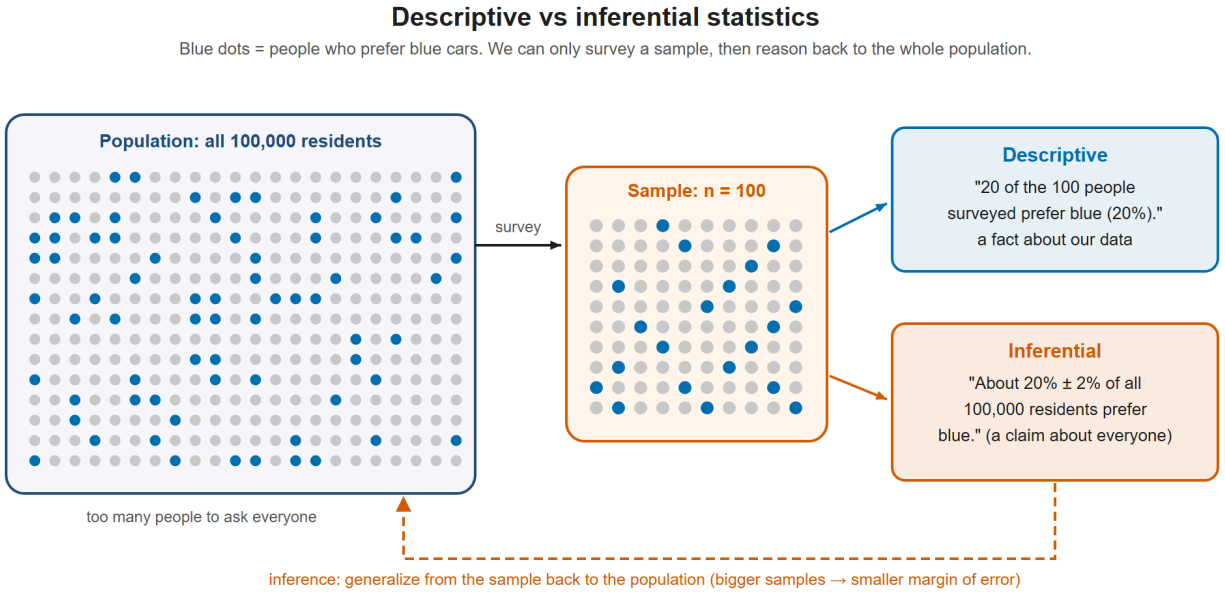
Inferential statistics

Inferential statistics uses **sample data** to make an **inference**, or draw a conclusion, about a **population**. It also uses probability to say **how confident** we can be in that conclusion.

Example - who likes blue cars?

City XYZ has **100,000** residents. We want to know what share of them prefer blue cars.

1. Asking all 100,000 people is not practical, so we survey a **sample of 100** people.
2. **20 out of 100** say they prefer blue cars, so **20%** of the sample. → This is a fact about the data we collected: **descriptive statistics**.
3. We then say: "About 20% of *all* residents prefer blue cars, within a 2% margin of error." → This is a conclusion about people we did not ask: **inferential statistics**.
4. A **confidence interval** is also inferential: "We are 95% confident that 20% ± 2% of residents prefer blue cars."



Describing the 100 people we surveyed is descriptive statistics. Using them to say something about all 100,000 residents is inferential statistics.

A bigger sample gives a better estimate

Suppose we survey **1,000** people instead and **21%** say yes. The margin of error shrinks to about **±1%**. A larger sample gives a more precise picture of the population.

Sample size	Sample result	Margin of error
100	20%	± 2% (as stated in the video)
1,000	21%	± 1% (as stated in the video)

> **Note:** the margins of error here are illustrative numbers from the video. Chapter 9 shows how to actually calculate a margin of error for a proportion. With the standard formula, a sample of 100 at 95% confidence gives a margin of about ±8%. The idea is still the same: **more data → smaller margin of error**.

Key takeaways

- **Descriptive** = summarize what you observed (graphs, tables, mean, median, mode, range, SD).
- **Inferential** = generalize from a **sample** to the **population**, with a stated level of confidence.
- Reporting "20 of 100 people said yes" is descriptive. Saying "20% of the city agrees" is inferential.
- Bigger samples give more precise inferences (smaller margin of error).